CAREFLOW

AI Powered Intelligent Hospital Patient Flow Management System

REVIEW 1 · PROBLEM IDENTIFICATION AND RESEARCH

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Guide: [GUIDE NAME] · CSE (Data Science) · Batch A10

Evidence-based documentation · Synthetic model data · Real running-system captures

NAVIGATION

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Abstract

•Hospitals coordinate registration, queues, consultation, laboratory work and beds through changing shared state.

•CareFlow creates digital tokens, tracks visits, exposes laboratory and capacity state, and publishes live role views.

•Two models assist triage priority and wait estimation; both are evaluated only on synthetic data.

•The project demonstrates software integration—not clinical validity or measured wait reduction.

7 Departments

5+1 Staff roles + public

2 Saved ML models

Base Paper: Alleviate

- IEEE Access, 2025; DOI 10.1109/ACCESS.2025.3647089.
- Flutter/Dart + Flask + Firebase integrated appointments, prescriptions, maps, clinic dashboards and personalised guidance.
- Reported 97.24% precision belongs to specialist recommendation—not hospital triage.
- CareFlow adopts the integrated dashboard idea but focuses on internal patient flow and resource state.

97.24% Paper's recommendation precision

Different Task and dataset

Literature Survey

- Access and location: hospital discovery, routing and appointment support.
- Prediction: disease/specialist recommendation from symptoms.
- Personalisation: diet, exercise and chronic-care guidance.
- Conversational support: NLP/chatbots for health queries.
- Waiting-time research motivates prediction, but institutions and outcome definitions differ.

19 References retained

4 Research themes

Supporting Evidence

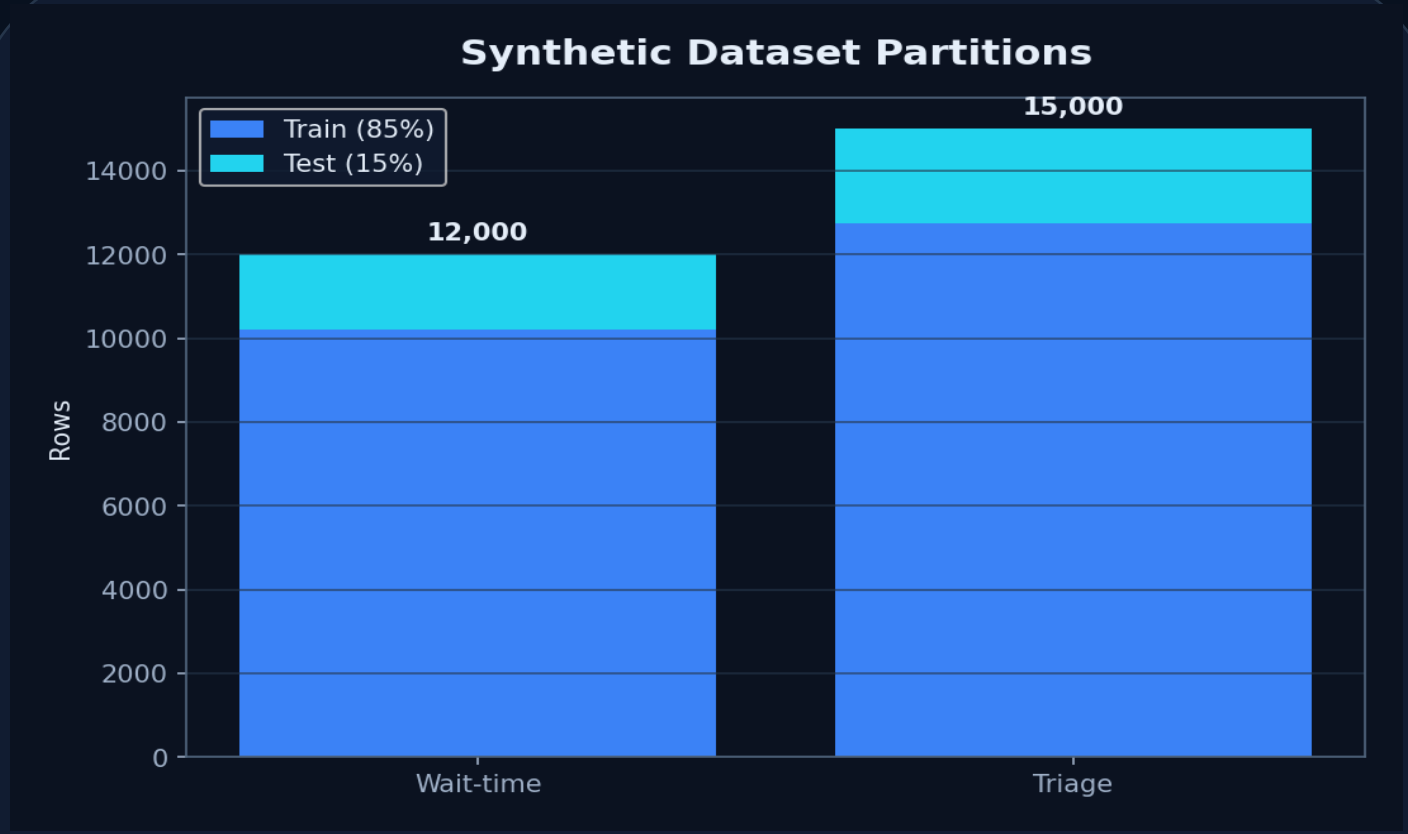


Figure: project dataset partitions

- 12,000 wait rows
- 15,000 triage rows
- 85/15 splits
- NumPy seed 42
- No real patient records

PROBLEM

Existing System

- Registration and billing may be digital while internal coordination remains fragmented.
- Name calling and simple tokens do not preserve priority or department movement.
- Laboratory work and bed availability may rely on calls or registers.
- Patients and staff lack one consistent current-state view.

Manual Queue coordination

Fragmented Department state

Limitations of Existing Approaches

- Appointment apps stop before the in-hospital journey.
- Prediction-only studies do not provide operational actions or resource state.
- Stale bed data delays admission decisions.
- Unexplained scores can hide class-specific safety failures.
- Disconnected systems create duplicate entry and weak traceability.

No	**Unified visit state**
No	**Live capacity view**

Research Gap

- Integrate token, priority, visit status, movement, laboratory work and bed state.
- Give each authorised role a focused action surface over the same data.
- Publish public queue progress without exposing patient identity.
- Separate synthetic-model evidence from clinical or operational impact claims.

One Operational source of truth

Explicit Evidence boundary

Problem Statement

- Design and implement a role-based system that coordinates patients across OPD, emergency, laboratory, pharmacy, ward and ICU.
- Preserve priority and current state, expose resource availability, and provide reproducible prediction assistance.
- Report limitations honestly where real data, validation or deployment evidence is absent.

Input — Patient + vitals + department

Output — Token + state + wait

Project Objectives

- Generate department tokens and priority-aware queues.
- Track visit state and explicit department movement.
- Coordinate laboratory orders and progress.
- Expose ward and ICU capacity in real time.
- Predict triage class and expected wait from documented features.
- Provide role dashboards plus a public waiting board.

6 Objectives

0 Clinical claims

Proposed Approach

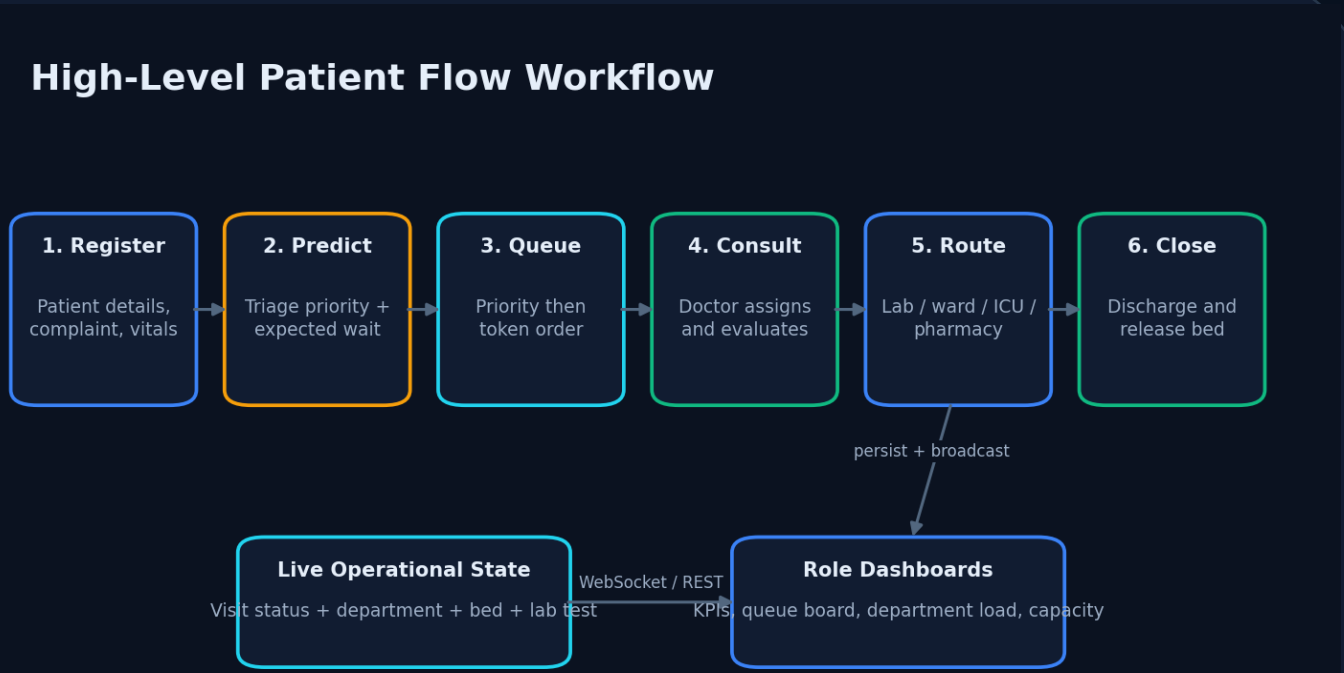


Figure: end-to-end workflow

- Register
- Predict
- Queue
- Consult
- Route
- Close

System Architecture

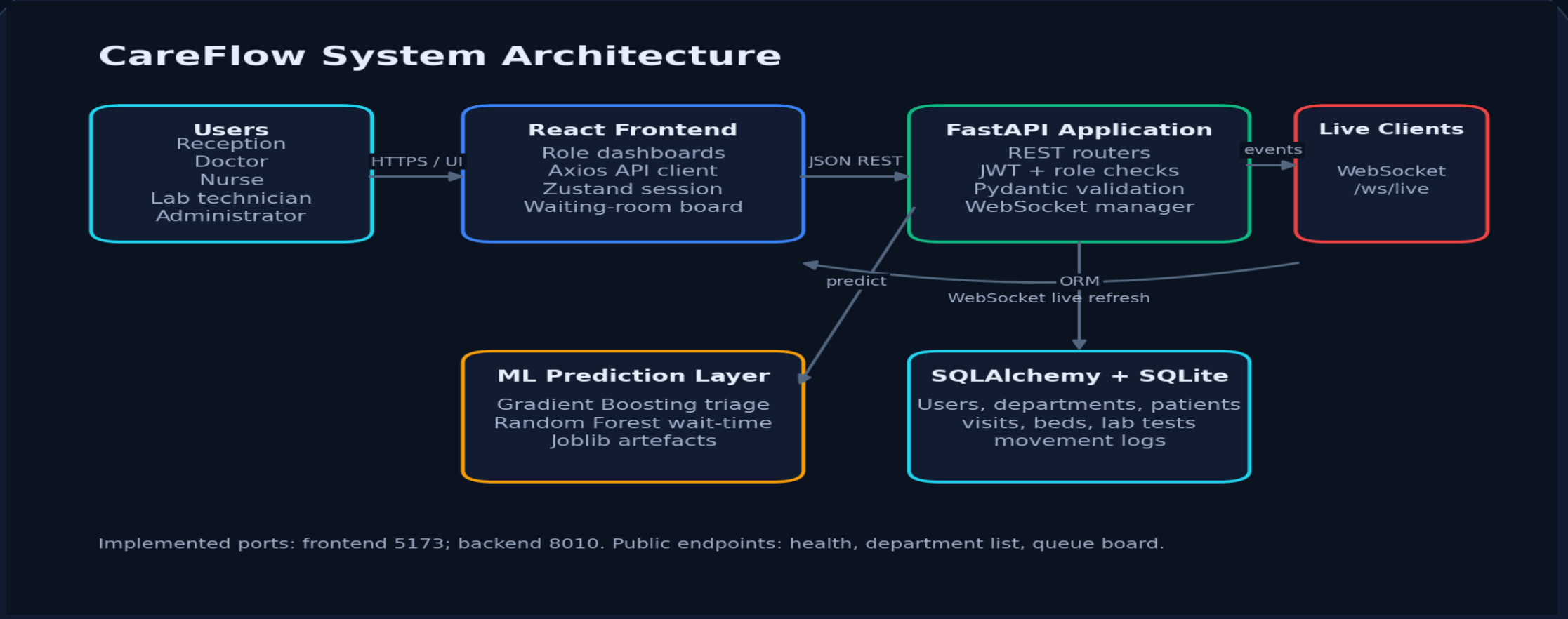


Figure: implemented application architecture

Expected Outcomes and Contributions

- A working multi-role prototype with populated operational screens.
- Reproducible synthetic model evaluation with class-specific reporting.
- Auditable queue ordering and movement evidence.
- Measured local API availability and latency.
- A clear list of unmeasured outcomes and production gaps.

150/150 Read smoke requests

13 Real screenshots

References

- [1] Saritha et al., Alleviate, IEEE Access, 2025, doi: 10.1109/ACCESS.2025.3647089.
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THANK YOU

Questions and suggestions are welcome

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